DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL
FINAL EXAMINATION-2018
Course Title: Discrete Mathematics
Course Code: CSE-1203
1st Year, 2nd Semester
Session: 2017-18

Time: 3 hours **Marks:** 60

Answer any five Questions from the followings. Parts of the same question should be answered consecutively.

1. a) Write these propositions using p, q, r and logical connectives. [5]
   i. Whenever you get a speeding ticket, you are driving over 65 miles per hour.
   ii. To take discrete mathematics you must have taken calculus or a course in computer science.
   iii. Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.
   iv. You will get an A in the class if and only if you either do every exercise in this book or you get an A in the final.
   v. You cannot ride the roller coaster if you are less than 4 feet tall unless you are older than 16 years old.

   b) Show that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology by using truth tables. [4]

   c) What rules of inference are used in these arguments? [3]
   "If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sun burn. Therefore, if I go swimming, then I will sun burn."

2. a) Define set with example. What is the power set of the empty set and the set $\{\emptyset\}$. [3]

   b) What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$? Find the relation $a < b$ and $a < c$, where $a \in A, b \in B$, and $c \in C$. [3]

   c) Use a membership table to show that $\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}$, where A, B and C are sets. [3]

   d) A given company has 1500 employees. Of those employees, 800 are computer science majors. 25% of those computer science majors are also mathematics majors. That group of computer science/math dual majors makes up one third of the total mathematics majors. How many employees have majors other than computer science and mathematics? [3]

3. a) Define domain, co-domain and range of a function. Let A{2,3,4} and B={a, b, c} and f={(2,a),(3,b),(4,b)} is a function from set A to B, find domain , co-domain and range of the function. [4]

   b) Explain bijective function with example. Determine whether $f(x) = x^2 + 1$ is a bijection from **Z** to **Z**. [3]

   c) Consider f, g and h, all the functions on the integers by $(n) = n^2$, $g(n) = n + 1$ and $h(n) = n - 1$. Determine [3]
   i) $hofog$
   ii) $gofoh$
   iii) $fogoh$

   d) Draw the graph of the function $f(x) = -x^3 - 5$ from **Z** to **Z**. [2]

4. a) Explain equivalence relation with example. Let A = {1, 2, 3, 4} and R = {(1,2), (1,1),(2,1),(2,2),(3,3),(4,4),(4,5),(5,4),(5,5)} be relation on A. Determine whether the relation R is equivalence relation. [4]

   b) Let R be a relation on set A={1, 2, 3, 4} defined by R={(1,1),(2,2),(3,3),(4,4),(4,3),(4,2),(4,1),(3,2),(3,1)} find the zero – one matrix and directed graph of relation R. [2]

   c) List the ordered pairs in the relations on {1, 2, 3, 4} corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order) and find the matrix representing R^{-1}. [3]

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

   d) Explain reflexive, symmetric and transitive closure of a relation with proper example. [3]

5. a) What are the terms a_1, a_3, a_5 and a_7 of the sequence $\{a_n\}$, where a_n equals $\lfloor n/2 \rfloor + \lceil n/2 \rceil$? [3]

   b) Compute value of the double sum: $\sum_{i=0}^{3}\sum_{j=0}^{8}(1)^i(-1)^j$. [3]

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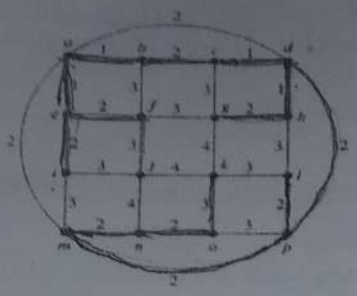
c) Find the minimal spanning tree and the minimum cost for the following figure [3]



1+1+1+1

d) Let $A = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}$, show that $(A^{-1})^3$ is the inverse of A^3. [3]

6. a) Explain recurrence relation with example. Write the recurrence relation of Fibonacci numbers and find the Fibonacci numbers $f2, f3, f4, f5$, and $f6$. [4]

b) Use mathematical induction to prove $1.2 + 3.4 + 5.6 + - - - - - - - - + (2n-1)2n = n(n+1)(4n-1)/3$ for all positive integers n. [4]

c) What are the quotient and remainder when -111 is divided by 11? [2]

d) For how many positive integer values of x is $\frac{65}{x}$ an integer. Determine all those values. [2]

7. a) A state issues automobile license plates that begin with two capital letters selected from a 26 letter alphabet, followed by four numerals selected from the digits 0 through 9, inclusive. Repeats are permitted. For example, one possible license plate combination is GF3352. How many unique license plate can be issued? [3]

b) Briefly explain the Pigeonhole principle with example. Show that if 9 books are to be kept in 4 shelves, there must be at least one self which contain at least 3 books. [3]

c) A bag contains 6 black chips numbered $1-6$, respectively, and 6 white chips numbered $1-6$, respectively. If Pavel reaches into the bag of 12 chips and removes 2 chips, one after the other, without replacing them, what is the probability that he will pick black chip #3 and then white chip #3? [2]

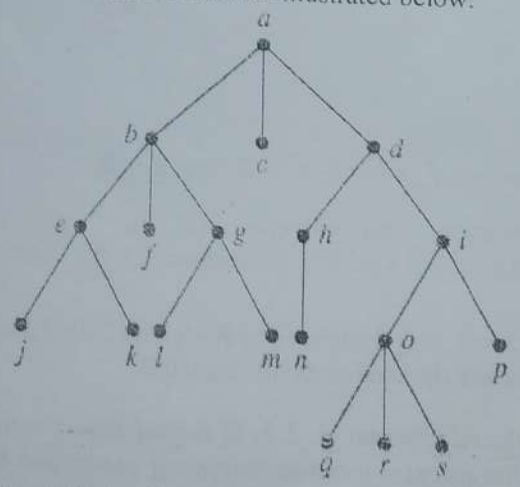
d) Explain conditional probability with example. Suppose that one person in 1,000 people has a genetic disease. There is an excellent test for the disease; 99.9% of people with the disease also test positive and only 0.02% who do not have the disease may test positive. What is the probability that someone who tests positive has the disease? [4]

8. a) Briefly explain $K_{3,6}$ and W_6 graphs with proper figures. How many edges does the W_6 graph have? [4]

b) Briefly explain adjacency list, adjacency matrix and incidence matrix with proper examples. [3]

c) Explain isomorphic graph with proper example. [2]

d) Answer the following questions about the rooted tree illustrated below. [3]



i. Is this a full $m-ary$ tree for some positive integer m?

ii. What is the level of each vertex of the above rooted tree?

iii. Draw the subtree of the above tree that is rooted at b.

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